How the BFKL term comes out of Group II, Row IV

1 Starting point

Notation:

- $d_\perp = 2-\epsilon$, $a \equiv k^+/\vee$, $X \equiv y'-y$, $c \equiv \frac{g^2 N_c}{4\pi^2}$, $\mathcal{K}(r) \equiv \frac{r}{r^2}$.
- $\mathcal{B} \equiv -\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{X^2\mu^2}{4} - 1 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ln(X^2\hat{\mu}^2)$, with $\hat{\mu}^2 = \mu^2 e^{2\gamma_E}/4$.
- $F(\xi)$ is the LO with $\mathcal{B}$ inside the integral and the gluon momentum rescaled by $1/\xi$:

$$F(\xi) \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-i\mathbf{k}\cdot(y'-y)/\xi}\,\mathcal{B}(y'-y)\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') \times \big[U^{e'c}(y') - U^{e'c}(x')\big]\big[U^{ec}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \quad (1.1)$$

- $F(1) = \frac{d^3 N_{\rm LO}}{d^3 k}[\mathcal{B}]$.

Row IV (your three lines, your p^+ prescription):

$$\begin{aligned}
\text{Row IV} = c\int_a^1 \frac{d\xi}{\xi^2}\,\xi(1-\xi)\,F(\xi) &\quad \text{(A)}\\
+ c\int_a^1 \frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\,F(\xi) &\quad \text{(B)}\\
+ c\int_a^1 d\xi\,\frac{F(\xi)-F(1)}{\xi(1-\xi)} &\quad \text{(C)}\\
+ c\Big[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+}-1\Big)\Big]F(1) &\quad \text{(D)}\\
+ c\int_a^{\frac{k^+}{k^+ + \Lambda}} d\xi\,\frac{1-\xi}{\xi}\,F^{\rm lp}(\xi) &\quad \text{(E)}
\end{aligned} \quad (1.2)$$

Question: which of these produce $\log\frac{\vee}{k^+}$ (i.e. $\log\frac{1}{a}$), and with which Wilson lines?

2 Step 1: how $F(\xi)$ behaves as $\xi \to 0$

Every lower limit $a = k^+/\vee \to 0$, so a $\log\vee$ can only come from the region $\xi \to 0$. We therefore need $F(\xi)$ at small ξ. There the phase $e^{-i\mathbf{k}\cdot X/\xi}$ oscillates fast.

1a. Separate the y' integral. Write

$$F(\xi) = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_y\int d^2y'\,e^{-i\mathbf{k}\cdot(y'-y)/\xi}\,\mathcal{B}(y'-y)\,\Phi_y(y'), \quad (2.1)$$

where

$$\Phi_y(y') \equiv \int_{x,x'} \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\big[U^{e'c}(y') - U^{e'c}(x')\big]\big[U^{ec}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x) \quad (2.2)$$

is a smooth function of y' (the x' integral smears the $\mathcal{K}(x'-y')$ singularity).

1b. The constant part of $\mathcal{B}$ drops out. $-\frac{2}{\epsilon}$ and $-\ln\hat{\mu}^2$ do not depend on X. They multiply

$$\int d^2y'\,e^{-i\mathbf{k}\cdot(y'-y)/\xi}\,\Phi_y(y'), \quad (2.3)$$

the Fourier transform of a smooth function at the large momentum $\mathbf{k}/\xi$. It falls faster than any power of ξ; for $\Phi_y \to \Phi_y(y)$ it is $(2\pi\xi)^2\delta^2(\mathbf{k})\Phi_y(y) = 0$ at $\mathbf{k} \neq 0$. So the pole and μ give nothing at small ξ.

1c. The $\ln X^2$ part. From $\nabla^2 \ln X^2 = 4\pi\delta^2(X)$, Fourier transforming at $\mathbf{q} = \mathbf{k}/\xi$ gives $-\mathbf{q}^2 \widetilde{\ln X^2} = 4\pi$, i.e.

$$\int d^2X\, e^{-i\mathbf{k}\cdot X/\xi}\, \ln X^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2}. \tag{2.4}$$

- This is a power, not exponentially small, because $\ln X^2$ is singular at $X = 0$.
- So at small ξ only $X \to 0$ matters: expand $\Phi_y(y') = \Phi_y(y) + X \cdot \nabla\Phi_y + \dots$. The higher terms are suppressed by further powers of ξ.
- The coefficient is $-\frac{d_\perp}{2} \to -1$, since no pole multiplies it.

Therefore

$$\int d^2y'\, e^{-i\mathbf{k}\cdot(y'-y)/\xi}\, \mathcal{B}(y'-y)\, \Phi_y(y') = +\frac{4\pi\xi^2}{\mathbf{k}^2}\, \Phi_y(y)\, \big[1 + O(\xi)\big]. \tag{2.5}$$

1d. Set $y' = y \equiv z$.

$$\Phi_z(z) = \int_{x,x'} \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \tag{2.6}$$

Both gluon points are now at the same z, in the amplitude and in the conjugate. Result of Step 1:

$$\boxed{F(\xi) \xrightarrow{\xi\to 0} \frac{4\pi\xi^2}{\mathbf{k}^2}\, \frac{d^3N_{\rm LO}}{d^3k}^{y}\, \big[1 + O(\xi)\big], \qquad \frac{d^3N_{\rm LO}}{d^3k}^{y} \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_z \Phi_z(z).} \tag{2.7}$$

Checked numerically in `smallxi.py`: the ratio to the leading term is 1.008, 1.002, 0.99994, 0.99979 at $\xi = 0.2, 0.1, 0.05, 0.03$. In the same way $F^{\rm lp}(\xi) = O(\xi^2)$.

3 Step 2: which terms give $\log\frac{\vee}{k^+}$

Insert Eq. (2.7) into each line and look at the integrand as $\xi \to 0$.

line	integrand at $\xi \to 0$	$\log\frac{\vee}{k^+}$?
(A) $\frac{\xi(1-\xi)}{\xi^2}F$	$\frac{1}{\xi}\cdot\xi^2 = \xi$	no (integrable)
(B) $\frac{1-\xi}{\xi^3}F$	$\frac{1}{\xi^3}\cdot\xi^2 = \frac{1}{\xi}$	**yes**
(C), $F(\xi)$ part: $\frac{F}{\xi(1-\xi)}$	$\frac{1}{\xi}\cdot\xi^2 = \xi$	no
(C), $-F(1)$ part	$-\frac{F(1)}{\xi(1-\xi)}$	$-F(1)\log(\frac{\vee}{k^+} - 1)$
(D), $\log(\frac{\vee}{k^+} - 1)F(1)$	–	$+F(1)\log(\frac{\vee}{k^+} - 1)$
(E) $\frac{1-\xi}{\xi}F^{\rm lp}$	$\frac{1}{\xi}\cdot\xi^2 = \xi$	no

The two logs with the LO Wilson lines cancel. (C) is finite at $\xi = 1$ only as the combination $F(\xi) - F(1)$, so split its range at $\xi = \frac{1}{2}$, where every piece is finite:

$$\text{(C)} = c\int_a^{1/2} d\xi\, \frac{F(\xi)}{\xi(1-\xi)} - c\,F(1)\int_a^{1/2}\frac{d\xi}{\xi(1-\xi)} + c\int_{1/2}^1 d\xi\, \frac{F(\xi)-F(1)}{\xi(1-\xi)}. \tag{3.1}$$

- The last integral does not depend on a.
- The first has integrand $\sim \xi$ at $\xi \to 0$: no $\log\vee$.
- The middle one is exact: $\int_a^{1/2}\frac{d\xi}{\xi(1-\xi)} = \Big[\log\frac{\xi}{1-\xi}\Big]_a^{1/2} = \log\frac{1-a}{a} = \log\Big(\frac{\vee}{k^+} - 1\Big)$.

So the $-F(1)$ part of (C) is exactly $-c\,F(1)\log(\frac{\vee}{k^+} - 1)$. This is equal and opposite to the $\log(\frac{\vee}{k^+} - 1)$ in (D). So there is no $\log\frac{\vee}{k^+}$ multiplying $\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]$. That is why the $\log(\frac{\vee}{k^+} - 1)$ of line 2 cannot be your BFKL term: it does not survive.

The only surviving log: line (B). Using Eq. (2.7),

$$(\mathrm{B}) = c\int_a^1 d\xi\,\frac{1-\xi}{\xi^3}\,\frac{4\pi\xi^2}{\mathbf{k}^2}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}^{y}\big[1+O(\xi)\big] = \frac{4\pi c}{\mathbf{k}^2}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}^{y}\int_a^1\frac{d\xi}{\xi} + \big(\text{no }\log\vee\big) = \log\frac{\vee}{k^+}\,\frac{g^2N_c}{\pi\mathbf{k}^2}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}^{y} + \dots \quad (3.2)$$

using $\frac{4\pi c}{\mathbf{k}^2} = \frac{g^2N_c}{\pi\mathbf{k}^2}$.

- (B) is the $\frac{1-\xi}{\xi}$ term, the $\frac{1}{\xi}$ (small-x) pole of P_{gg}.
- $\xi\to 0$ means $p^+ = k^+\frac{1-\xi}{\xi} \gg k^+$: the other gluon is much faster than the measured one.
- The fast gluon is produced at z; the measured, softer gluon is emitted from it.

4 Step 3: write it as H_4 (your Wilson-line structure)

To put back the emission of the measured gluon in coordinate space, use

$$\int_w e^{i\mathbf{k}\cdot w}\,\mathcal{K}^i(w-z) = \frac{2\pi i\,k^i}{\mathbf{k}^2}e^{i\mathbf{k}\cdot z} \quad\Rightarrow\quad \int_{w,w'} e^{-i\mathbf{k}\cdot(w'-w)}\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z) = \frac{4\pi^2}{\mathbf{k}^2}. \quad (4.1)$$

This is finite: integrating over w, w' gives no $\log\mu$. So $\frac{1}{\mathbf{k}^2} = \frac{1}{4\pi^2}\int_{w,w'} e^{-i\mathbf{k}\cdot(w'-w)}\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$, and

$$\frac{g^2N_c}{\pi\mathbf{k}^2}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}^{y} = \frac{1}{(2\pi)^3}\,\frac{g^2N_c}{\pi}\,\frac{g^2}{2\pi^2k^+}\,\frac{1}{4\pi^2}\int_{x,x',z,w,w'}\cdots = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5k^+}\int_{x,x',z,w,w'}\cdots \equiv H_4. \quad (4.2)$$

5 Result: the BFKL term of Group II, Row IV

$$\boxed{\mathrm{BFKL}\big|_{\mathrm{II,IV}} = \log\frac{\vee}{k^+}\,H_4} \quad (5.1)$$

$$H_4 = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-i\mathbf{k}\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\Big\{U^{e'c}(x')\,U^{ce}(x)\,\rho^{e'}(x')\rho^e(x) + \rho^e(x')\rho^e(x) - U^{e'c}(z)\,U^{ce}(x)\,\rho^{e'}(x')\rho^e(x) - U^{e'c}(x')\,U^{ec}(z)\,\rho^{e'}(x')\rho^e(x)\Big\}. \quad (5.2)$$

The braces are $[U^{e'c}(z)-U^{e'c}(x')][U^{ec}(z)-U^{ce}(x)]\rho^{e'}(x')\rho^e(x)$ expanded, with $U^{e'c}(z)U^{ec}(z) = \delta^{e'e}$ (adjoint, real orthogonal) for the second line.

Which terms are associated with it.

term of Row IV	role in BFKL
(B): $c\int_a^1\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}F(\xi)$	**the source**: its $\xi\to0$ end gives $\log\frac{\vee}{k^+}H_4$
(C) $-F(1)$ part and (D) $\log(\frac{\vee}{k^+}-1)$	cancel each other; no BFKL
(A), (C) $F(\xi)$ part, (E)	integrable at $\xi\to0$; no BFKL
(D) $\log\frac{k^+}{\Lambda}$	JIMWLK (not BFKL)

Cross-check without any expansion. $\frac{d\,\text{Row IV}}{d\log\vee} = cF(a)\big[\frac{1-a}{a^2}+(1-a)+\frac{1}{1-a}\big] + c(1-a)F^{\mathrm{lp}}(a)$ exactly; the $\pm F(1)$ of (C) and (D) cancel here too. With Eq. (2.7) this is $H_4 + O(\frac{k^+}{\vee})$. Checked numerically in `dlogvee.py` (to 10^{-11}).